Homework 4 – Regression Consult

When people make decisions, their choices are often swayed by how information is presented. This is often described as "the framing effect." One example of this, is how commercials describe new medications. As an example, Nereus (tradipitant), a new drug for motion-induced vomiting like sea-sickness, is marketed with a **positive frame** as reducing vomiting incidence from 44% to 18% (a reduction of 60%). This marketing makes it seem more effective than a **negative frame** that says just 18% of patients still experienced vomiting, even though both statements are true.

Xu (2024) aimed to evaluate whether older adults are more or less vulnerable to framing than younger adults. They collected a sample of 696 adults aged 18 to 71 and exposed them to three different scenarios

Scenario 1: Rate Drug Efficacy (1 = very bad, 6 = very good)

- **Positive Frame:** 70 out of 100 patients felt better
- **Negative Frame:** 30 out of 100 patients did not feel better

Scenario 2: Likelihood to stop drinking cocoa (1 = surely stop, 6 = surely continue)

- **Positive Frame:** Stopping drinking hot cocoa reduces cadmium and cancer risk.
- **Negative Frame:** Continuing to drink hot cocoa increases cadmium and cancer risk.

Scenario 3: Preference for chance choice (1 = surely safe option, 6 = surely risky)

- **Positive Frame:** Choice between 200 saved vs 1/3 chance all 600 are saved.
- **Negative Frame:** Choice between 400 die vs 2/3 chance all 600 die.

In each scenario, participants were randomly assigned a positive or negative frame. They rated their response on a 1-6 scale. The researchers want to build three models, one for each scenario adjusting for simultaneous linear changes in demographic variables (gender, ethnicity, education level, political affiliation).

They want to evaluate whether, as adults age, the way choices are worded have more influence on their judgments and ultimate choices. If there is a change, are those changes driven by positive or negative framings? At what age does this effect onset?

Your Goals:

- Determine the correct analysis strategy and conduct the necessary analyses.
  - Hint 1: Ensure to assess and communicate results about the model fit.
  - Hint 2: Ensure to think about whether or which interactions need to be included.
  - Hint 3: You will need run various post hoc procedures (e.g., `emmeans`, `emtrends` or `johnson_neyman`).
  - Hint 4: Looking at the paper may be helpful. They do include `R` code, but I expect you to do a much better job using the techniques we discussed (they do somethings differently)
- Write a clear, concise, complete, and correct report conveying an answer to their research question.

The data are in `HW4Data.csv` but they will require some manual and `tidyverse` cleaning.

1 Loading and Cleaning the Data

```r
# Load Data
dat <- read_csv("HW4data.csv",
                col_names = F,
                col_types = cols(.default = col_character()))

# Clean Data
dat <- dat |>
  slice(1:698) |>      # rows 699-844 are empty rows
  select(X1:X32)       # X33-X52 are empty columns

names(dat) <- str_trim(as.character(dat[2, ]))  # row 2 becomes the column names

dat <- dat |>
  slice(-(1:2)) |>    # remove the question-text row and the names row
  rename(duration = duratin) |>
  type_convert(col_types = cols()) |>
  select(-c(confirmation, Q_RecaptchaScore, userLanguage, distributionChannel, startDate, endDate))

# Clean age attention check
dat <- dat |>
  mutate(band   = stri_trans_nfkc(ageattentioncheck),   # unbolding
         age_ok = is.na(band) |                         # early respondents weren't asked
                  between(age, parse_number(band), parse_number(str_remove(band, ".*-"))))

dat <- dat |>
  filter(age_ok, is.na(attentioncheck) | attentioncheck == 1) |>
  select(-c(ageattentioncheck, attentioncheck, band, age_ok))

# Remove "prefer not to say" answers / fix wordings
framing <- dat |>
  mutate(across(c(gender, ethnicity, politics, education), ~ na_if(.x, "Prefer not to say"))) |>
  drop_na(gender, ethnicity, politics, education) |>
  transmute(
    id = responseID,
    age,
    gender    = factor(gender),
    ethnicity = factor(ethnicity),
    politics  = factor(politics),
    education = factor(education,
                       levels = c("Less than high school", "High school graduate",
                                  "Some college", "2 year degree", "4 year degree",
                                  "Professional degree", "Doctorate")),
    frame_attribute   = factor(if_else(treatment1 == "pos1", "Positive", "Negative"),
                           levels = c("Negative", "Positive")),
    frame_goal  = factor(if_else(treatment2 == "pos2", "Positive", "Negative"),
                           levels = c("Negative", "Positive")),
    frame_risky = factor(if_else(treatment3 == "pos3", "Positive", "Negative"),
                           levels = c("Negative", "Positive")),
    resp_attribute   = pick1,
    resp_goal  = pick2,
    resp_risky = pick3
  )

# for plotting later
```

```r
framing_long <- framing |>
  pivot_longer(c(starts_with("frame_"), starts_with("resp_")),
               names_to = c(".value", "scenario"), names_sep = "_") |>
  mutate(scenario = factor(scenario,
                           levels = c("attribute", "goal", "risky"),
                           labels = c("Attribute (drug)",
                                      "Goal (cocoa)",
                                      "Risky choice (disease)")))
```

When I loaded the .csv file, there were extra columns and rows that were empty and needed to be removed. Additionally the first two rows were the full question followed by a shortened name for the column, so I made the shortened name the column header and removed the row containing the survey question, and removed columns that wouldn't be useful to our analysis, including reCAPTCHA score, confirmation, language, distribution style, and start and end date. Next, like the paper, I removed participants who didn't pass the attention and age attention checks, then removed the columns. Lastly, I chose to remove patients who answered "prefer not to say" for a question (n = 10) and fixed some of the wordings of the framing columns so it was easier for me to understand.

2 Exploratory Data Analyses

```r
framing_long |>
  group_by(scenario, frame) |>
  summarize(n        = sum(!is.na(resp)),
            mean     = mean(resp, na.rm = TRUE),
            sd       = sd(resp, na.rm = TRUE),
            mean_age = mean(age),
            .groups  = "drop") |>
  kable(digits = 2,
        col.names = c("Scenario", "Frame", "n", "Mean rating", "SD", "Mean age"))
```

Table 1: Ratings (1–6) and participant age by scenario and assigned frame.

Scenario	Frame	n	Mean rating	SD	Mean age
Attribute (drug)	Negative	358	3.84	0.80	40.72
Attribute (drug)	Positive	320	4.49	0.62	41.08
Goal (cocoa)	Negative	329	2.00	1.09	40.88
Goal (cocoa)	Positive	349	2.52	1.40	40.89
Risky choice (disease)	Negative	337	3.59	1.50	41.90
Risky choice (disease)	Positive	340	2.89	1.38	39.88

Random assignment worked as intended. In each scenario, the two frames have similar group sizes and similar mean ages (Table 1).

Attribute: The drug was rated as more effective under the positive frame than under the negative frame.

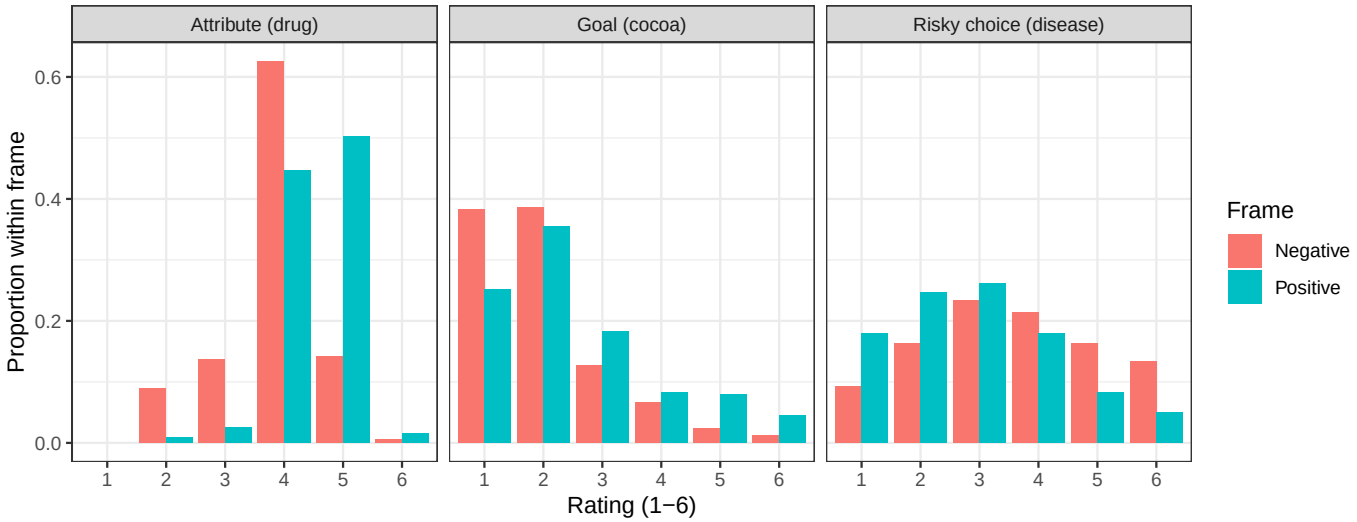
Goal: Under the negative frame, participants were more likely to say they would stop drinking cocoa.

Risky choice: The negative frame pushed participants toward the risky option.

```r
framing_long |>
  drop_na(resp) |>
  ggplot(aes(x = factor(resp), fill = frame)) +
  geom_bar(aes(y = after_stat(prop), group = frame), position = "dodge") +
  facet_wrap(~ scenario) +
```

```
  labs(x = "Rating (1-6)", y = "Proportion within frame", fill = "Frame") +
  theme_bw()
```

Figure 1: Distribution of ratings by frame within each scenario.



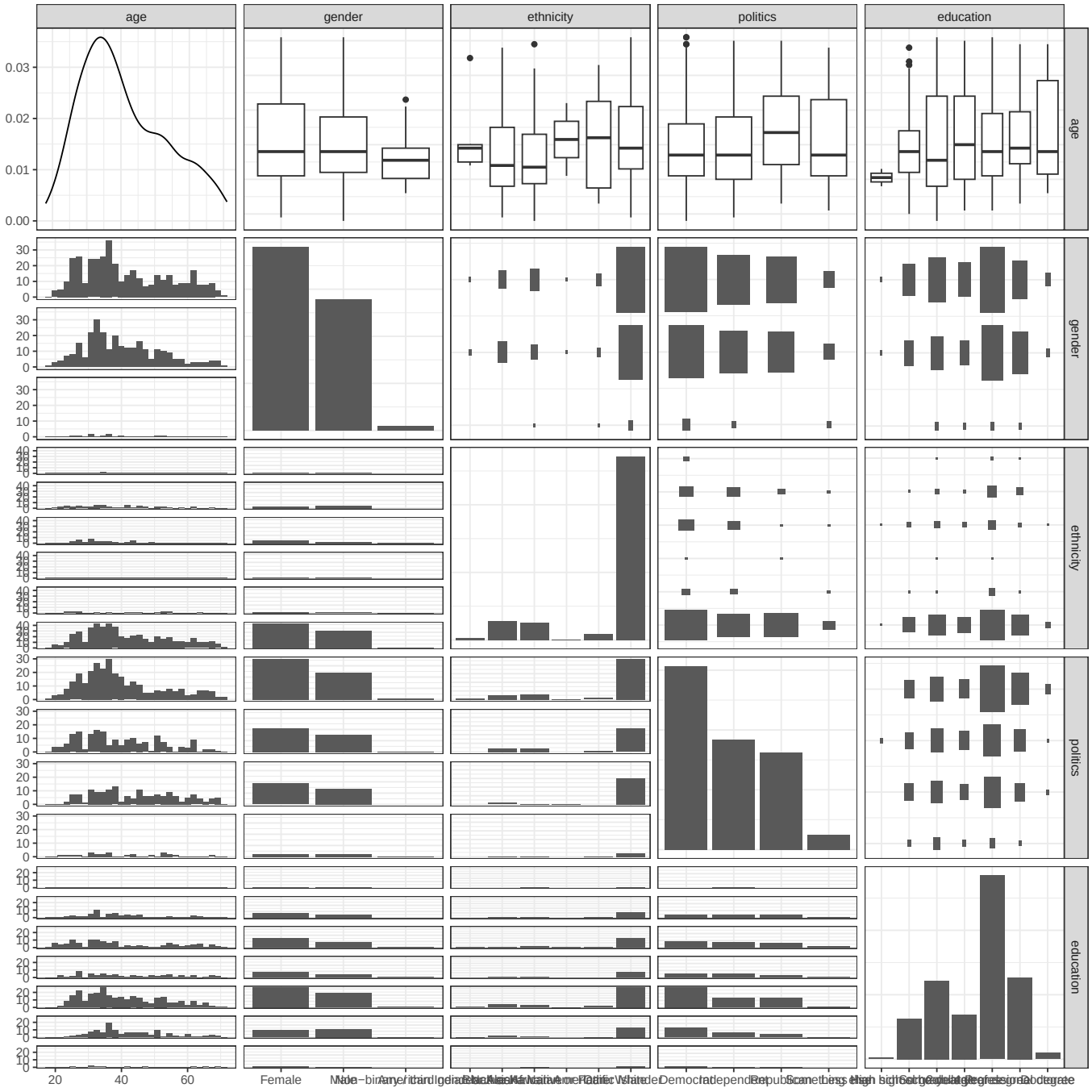
Distribution of positive and negative responses by framing subtype. Positive and negative attribute framing appears approximately normal. Positive and negative goal framing appears skewed right. Positive and negative risky choice framing appears approximately normal.

2.0.1 Pairwise plots

All three models share the same adjustment variables, so we check them once for strong relationships with each other.

```
ggpairs(framing,
        columns  = c("age", "gender", "ethnicity", "politics", "education"),
        progress = FALSE) +
  theme_bw()
```

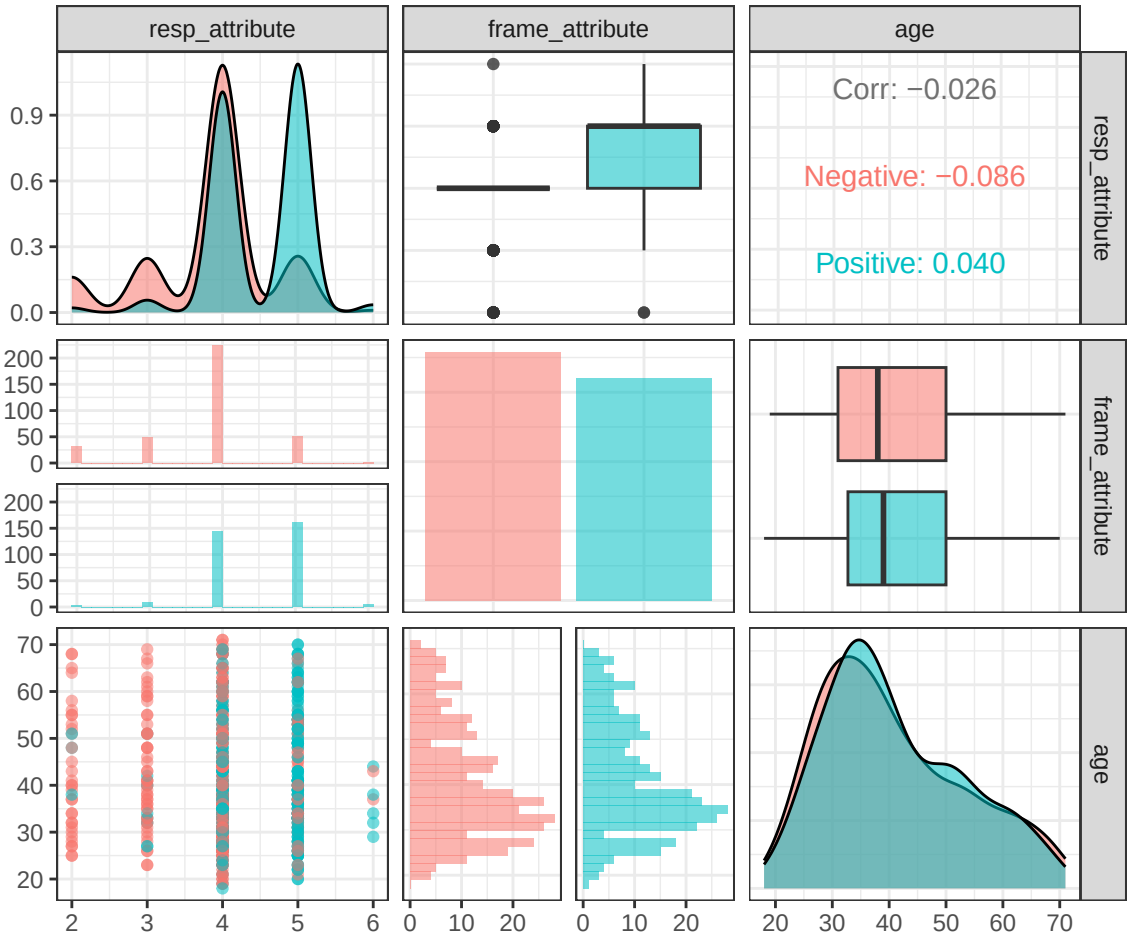
Figure 2: Pairwise plots of age and the demographic adjustment variables.



Pairwise plots to check collinearity between predictors. None of the demographic variables are strongly related to another (Figure 2). The clearest pattern is that Republicans are a few years older on average than the other political groups.

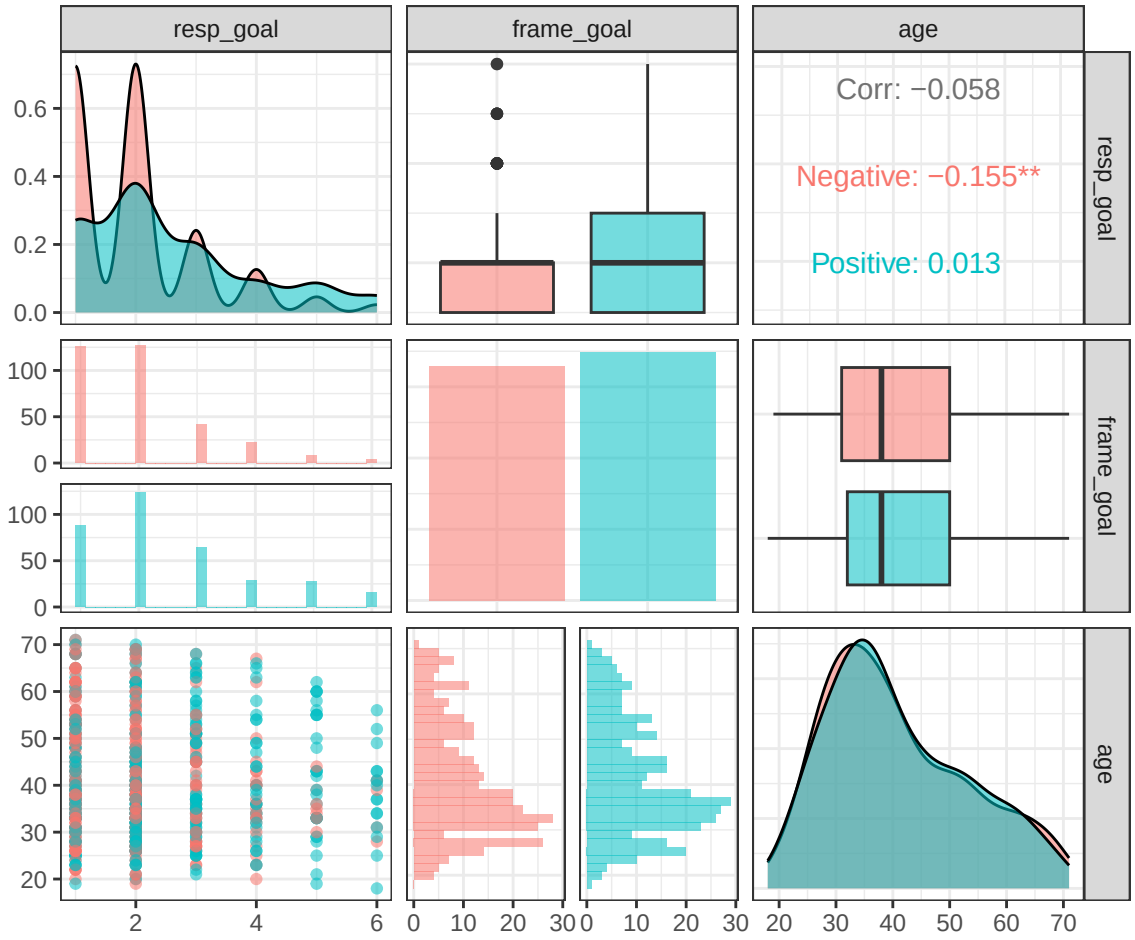
```
ggpairs(framing, columns = c("resp_attribute", "frame_attribute", "age"),
        mapping = aes(color = frame_attribute, alpha = 0.5), progress = FALSE) +
  theme_bw()
```

Figure 3: Pairwise plots for attribute framing (drug efficacy).



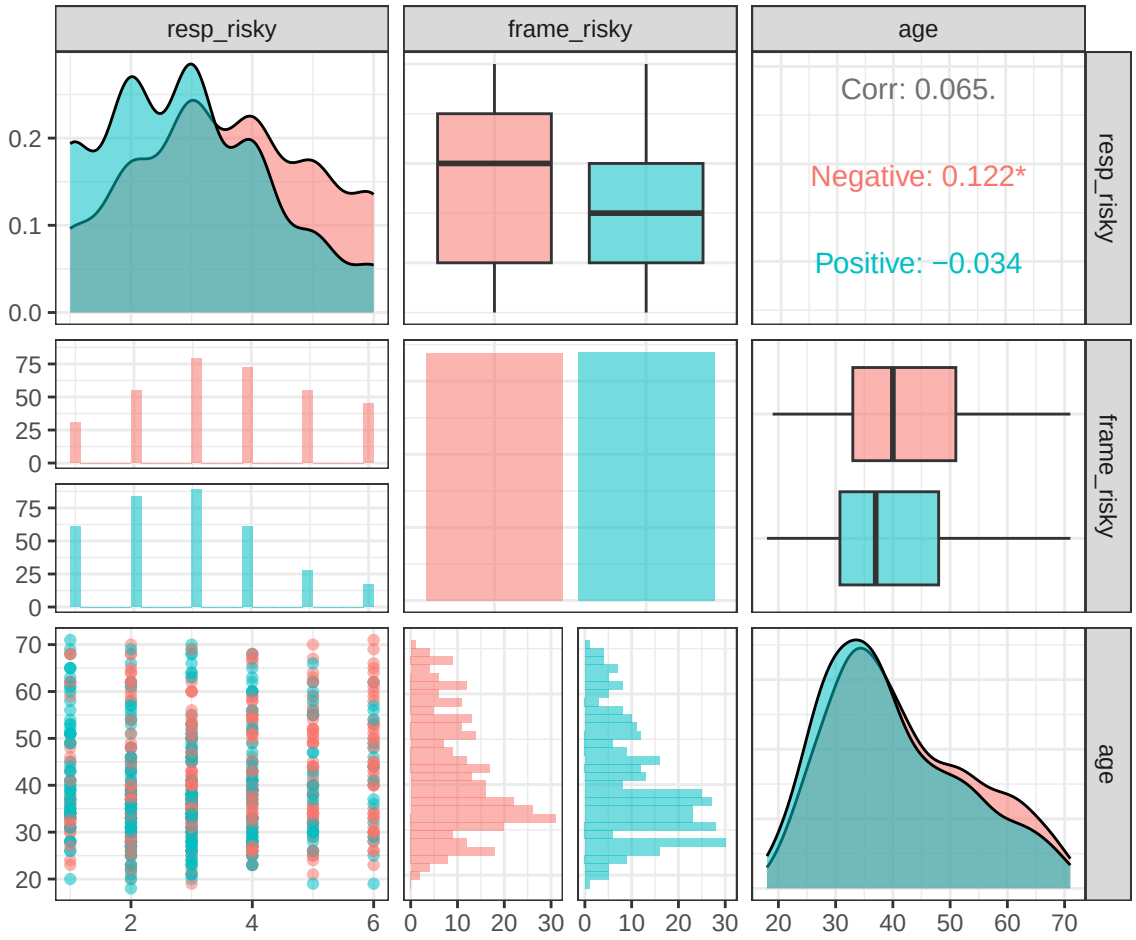

```
ggpairs(framing, columns = c("resp_goal", "frame_goal", "age"),
        mapping = aes(color = frame_goal, alpha = 0.5), progress = FALSE) +
  theme_bw()
```

Figure 4: Pairwise plots for goal framing (stop cocoa).




```
ggpairs(framing, columns = c("resp_risky", "frame_risky", "age"),
        mapping = aes(color = frame_risky, alpha = 0.5), progress = FALSE) +
  theme_bw()
```

Figure 5: Pairwise plots for risky-choice framing (disease).



Pairwise plots for each framing effect to check for linearity between framing and response, as well as collinearity between age and framing effect (minimized by randomization).

3 Regression Analyses

```
library(broom)          # tidy(), augment()
library(car)            # vif()
library(effectsize)     # F_to_omega2(), omega_squared(), interpret_omega_squared(), standardize_parameters(
library(emmeans)        # emmeans(), emtrends(), pairs()
library(interactions)   # johnson_neyman()
library(sandwich)       # vcovHC() for the robustness check
```

For each scenario {attribute, goal, risky}$, we fit

$$
\begin{aligned}
\hat{Y}_{i,s} = \beta_0 &+ \beta_1 I(\text{Frame}_i = \text{Pos}) + \beta_2\,\text{Age}_i + \beta_3\, I(\text{Frame}_i = \text{Pos}) \times \text{Age}_i \\
&+ \beta_4 I(\text{Gender}_i = \text{Male}) + \beta_5 I(\text{Gender}_i = \text{Non-binary}) \\
&+ \beta_6 I(\text{Eth}_i = \text{Asian}) + \beta_7 I(\text{Eth}_i = \text{Black}) + \beta_8 I(\text{Eth}_i = \text{Native Hawaiian/Pacific Islander}) \\
&+ \beta_9 I(\text{Eth}_i = \text{Other}) + \beta_{10} I(\text{Eth}_i = \text{White}) \\
&+ \beta_{11} I(\text{Pol}_i = \text{Independent}) + \beta_{12} I(\text{Pol}_i = \text{Republican}) + \beta_{13} I(\text{Pol}_i = \text{Something else}) \\
&+ \beta_{14} I(\text{Edu}_i = \text{High school}) + \beta_{15} I(\text{Edu}_i = \text{Some college}) + \beta_{16} I(\text{Edu}_i = \text{2-year}) \\
&+ \beta_{17} I(\text{Edu}_i = \text{4-year}) + \beta_{18} I(\text{Edu}_i = \text{Professional}) + \beta_{19} I(\text{Edu}_i = \text{Doctorate})
\end{aligned}
$$

where $\hat{Y}_i$ is participant i's predicted 1–6 rating in each scenario and $I(\cdot)$ equals 1 when the condition holds and 0 otherwise. The reference participant is a Female, American Indian or Alaska Native, Democrat with Less than a high school education who saw the negative frame. Each model has $p = 20$ coefficients.

```r
# Residual plots: Q-Q plot and residuals vs fitted values (+/- 2 and 3 sigma)
resid_plots <- function(mod) {
  d <- augment(mod)
  s <- sigma(mod)
  qq <- ggplot(d, aes(sample = .resid)) +
    stat_qq(alpha = 0.4) + stat_qq_line() +
    labs(x = "Theoretical Quantiles", y = "Observed Quantiles") +
    theme_bw()
  rf <- ggplot(d, aes(.fitted, .resid)) +
    geom_point(alpha = 0.4) +
    geom_hline(yintercept = 0, linetype = "dashed", color = "deeppink3") +
    geom_hline(yintercept = c(-2, 2) * s, linetype = "dotted", color = "steelblue") +
    geom_hline(yintercept = c(-3, 3) * s, linetype = "dotted", color = "darkorange") +
    labs(x = expression("Fitted Values (" * hat(Y) * ")"), y = "Residual") +
    theme_bw()
  list(qq = qq, rf = rf)
}

# Influence plots: leverage, Cook's distance, DFFITS, studentized residuals
influence_plots <- function(mod) {
  n <- nobs(mod); p <- length(coef(mod))
  d <- tibble(obs = seq_len(n),
              Leverage                = hatvalues(mod),
              `Cook's Distance`       = cooks.distance(mod),
              DFFITS                  = dffits(mod),
              `Studentized Residual`  = rstudent(mod)) |>
    pivot_longer(-obs, names_to = "measure") |>
    mutate(measure = fct_inorder(measure))
  cuts <- bind_rows(
    tibble(measure = "Leverage",            y = c(2, 3) * p / n,           level = c("Moderate", "High")),
    tibble(measure = "Cook's Distance",     y = qf(c(0.2, 0.5), p, n - p), level = c("Moderate", "High")),
    tibble(measure = "DFFITS",              y = c(-2, 2, -3, 3) * sqrt(p / n),
           level = rep(c("Moderate", "High"), each = 2)),
    tibble(measure = "Studentized Residual", y = c(-2, 2, -3, 3),
           level = rep(c("Moderate", "High"), each = 2))
  ) |> mutate(measure = factor(measure, levels = levels(d$measure)))
  ggplot(d, aes(obs, value)) +
    geom_point(size = 0.6, alpha = 0.6) +
    geom_hline(data = cuts, aes(yintercept = y, color = level), linetype = "dotted") +
    facet_wrap(~ measure, scales = "free_y") +
    scale_color_manual(values = c(Moderate = "steelblue", High = "darkorange")) +
    labs(x = "Observation Number", y = NULL, color = "Cutoff") +
```

```
    theme_bw()
}

# Coefficient table with standardized coefficients
coef_table <- function(mod) {
  tidy(mod) |>
    mutate(std = standardize_parameters(mod)$Std_Coefficient,
           p.value = if_else(p.value < 0.0001, "<0.0001", as.character(round(p.value, 4)))) |>
    kable(digits = 3, col.names = c("Term", "Estimate", "SE", "t", "p", "Std-Coef"))
}

# Partial omega-squared for each term in the (Type I) ANOVA
term_effects <- function(mod) {
  omega_squared(anova(mod)) |>
    as_tibble() |>
    mutate(Size = as.character(interpret_omega_squared(Omega2_partial, rules = "cohen1992"))) |>
    select(Parameter, Omega2_partial, CI_low, CI_high, Size) |>
    kable(digits = 3, col.names = c("Term", "Partial omega^2", "Lower", "Upper", "Size"))
}

# Estimated marginal means across age, by frame
emm_plot <- function(mod, frame, ylab) {
  emmeans(mod, specs = frame, by = "age", at = list(age = 18:71)) |>
    as.data.frame() |>
    ggplot(aes(age, emmean, color = .data[[frame]], fill = .data[[frame]])) +
    geom_ribbon(aes(ymin = lower.CL, ymax = upper.CL), alpha = 0.2, color = NA) +
    geom_line(linewidth = 1) +
    labs(x = "Age (years)", y = ylab, color = "Frame", fill = "Frame") +
    theme_bw()
}

# Key numbers for inline text
key_stats <- function(mod, frame) {
  s   <- summary(mod)
  int <- tidy(mod) |> filter(str_detect(term, ":age"))
  emt <- emtrends(mod, specs = frame, var = "age") |> test() |> as.data.frame()
  list(n = nobs(mod), r2 = s$r.squared, adj = s$adj.r.squared,
       F = s$fstatistic[1], df1 = s$fstatistic[2], df2 = s$fstatistic[3],
       Fp = pf(s$fstatistic[1], s$fstatistic[2], s$fstatistic[3], lower.tail = FALSE),
       int_b = int$estimate, int_p = int$p.value,
       neg_b = emt$age.trend[1], neg_p = emt$p.value[1],
       pos_b = emt$age.trend[2], pos_p = emt$p.value[2])
}
fp <- function(p) if (p < 0.001) "p < 0.001" else paste0("p = ", formatC(p, format = "f", digits = 3))
```

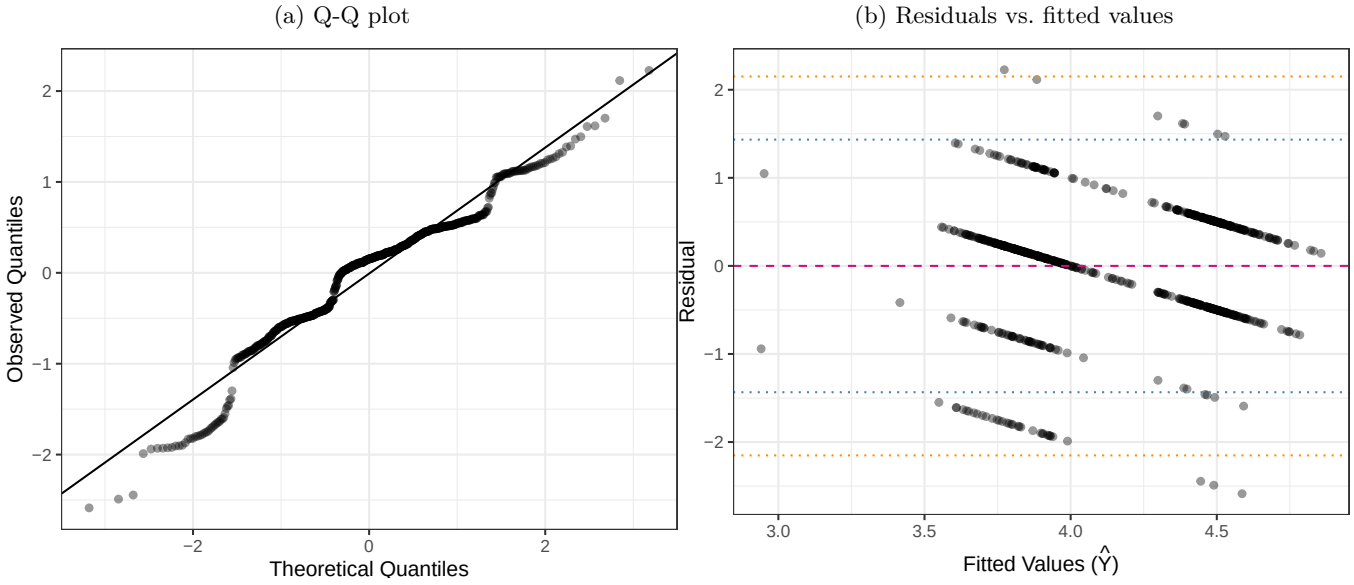
3.1 Rating Drug Efficacy

```
m_att <- lm(resp_attribute ~ gender + ethnicity + politics + education +
              frame_attribute * age,
            data = framing)
k_att <- key_stats(m_att, "frame_attribute")
```

Conditions.

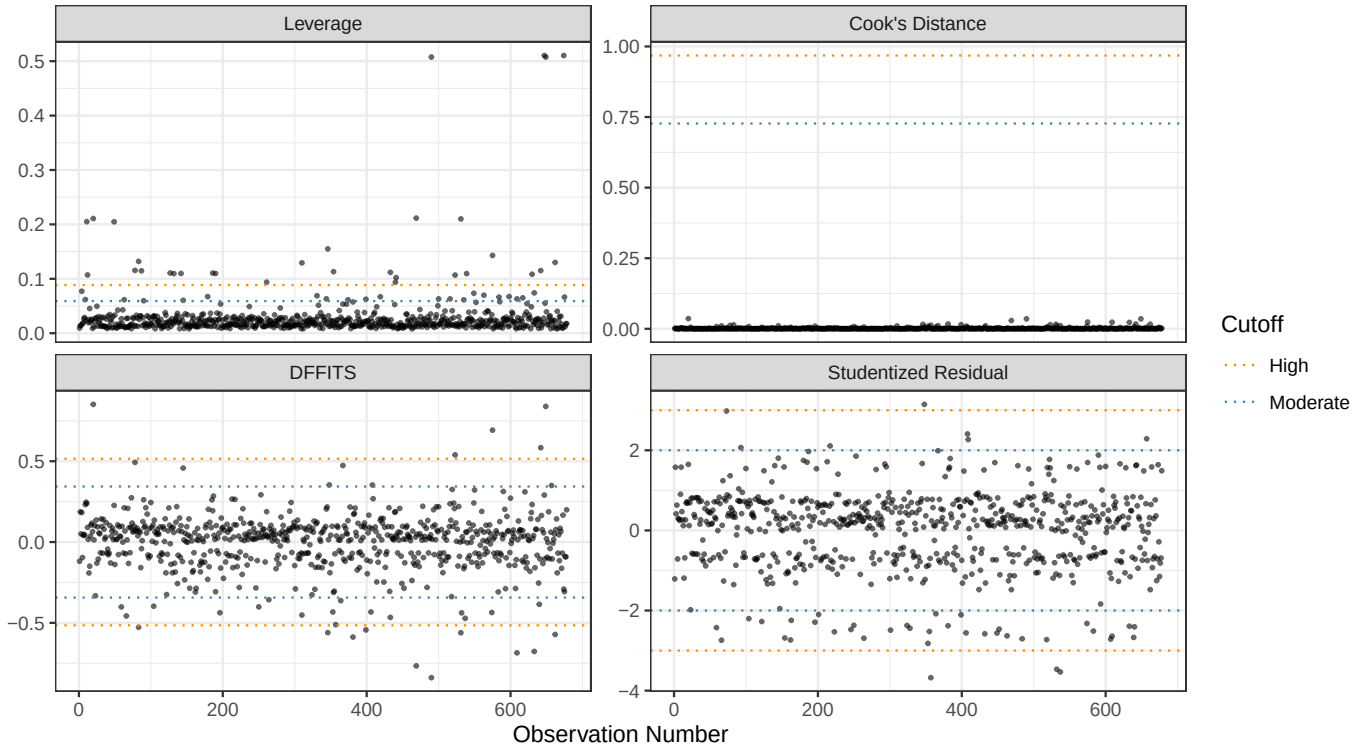
```
r <- resid_plots(m_att); r$qq; r$rf
```

Figure 6: Residual diagnostics for the attribute-framing model.




```
influence_plots(m_att)
```

Figure 7: Influence diagnostics for the attribute-framing model.




```
# Residual spread by frame (constant variance)
augment(m_att) |> group_by(frame_attribute) |> summarize(resid_sd = sd(.resid)) |> kable(digits = 3)
```

frame_attribute	resid_sd
Negative	0.778
Positive	0.619

```r
# VIF on the additive model (the interaction inflates VIF by construction)
vif(lm(resp_attribute ~ gender + ethnicity + politics + education + frame_attribute + age,
       data = framing))
```

```
                    GVIF Df GVIF^(1/(2*Df))
gender          1.069444  2        1.016926
ethnicity       1.151129  5        1.014174
politics        1.181429  3        1.028177
education       1.173558  6        1.013426
frame_attribute 1.030213  1        1.014994
age             1.059931  1        1.029530
```

- **Gaussian errors:** Because the response is a 1–6 rating with nearly all values at 4 or 5, the residuals fall in diagonal bands (Figure 6). Otherwise, the residuals look approximately Gaussian, with few outliers and influential points.
- **Constant variance:** The residuals are slightly more spread out under the negative frame than the positive frame.
- **Multicollinearity:** All $\text{GVIF}^{1/(2\,\text{Df})}$ values are close to 1, so there is no multicollinearity.

Overall F-test and effect size.

```r
anova(m_att) |> as.data.frame() |> kable(digits = 3)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
gender	2	0.305	0.153	0.297	0.743
ethnicity	5	5.881	1.176	2.288	0.045
politics	3	0.484	0.161	0.314	0.815
education	6	4.680	0.780	1.517	0.170
frame_attribute	1	71.293	71.293	138.655	0.000
age	1	0.155	0.155	0.302	0.583
frame_attribute:age	1	1.414	1.414	2.751	0.098
Residuals	658	338.330	0.514	NA	NA

```r
F_to_omega2(f        = k_att$F,
            df       = k_att$df1,
            df_error = k_att$df2)
```

```
Omega2 (partial) |       95% CI
--------------------------------
0.18             | [0.12, 1.00]

- One-sided CIs: upper bound fixed at [1.00].
```

```r
interpret_omega_squared(F_to_omega2(k_att$F, k_att$df1, k_att$df2)$Omega2_partial, rules = "cohen1992")
```

```
[1] "medium"
(Rules: cohen1992)
```

The model accounts for a statistically discernible share of the variation in efficacy ratings ($R^2 = 0.199$, adjusted $R^2 = 0.176$; $F(19, 658) = 8.62$, $p < 0.001$). Most of this comes from the frame (Table 2).

```
term_effects(m_att)
```

Table 2: Effect size (partial omega-squared) for each term, attribute framing.

Term	Partial omega^2	Lower	Upper	Size
gender	0.000	0.000	1	very small
ethnicity	0.009	0.000	1	very small
politics	0.000	0.000	1	very small
education	0.005	0.000	1	very small
frame_attribute	0.169	0.128	1	medium
age	0.000	0.000	1	very small
frame_attribute:age	0.003	0.000	1	very small

```
coef_table(m_att)
```

Table 3: Coefficients for the attribute-framing model.

Term	Estimate	SE	t	p	Std-Coef
(Intercept)	2.692	0.615	4.375	<0.0001	-2.077
genderMale	-0.007	0.057	-0.114	0.909	-0.008
genderNon-binary / third gender	0.105	0.233	0.449	0.6533	0.133
ethnicityAsian	0.799	0.337	2.371	0.018	1.011
ethnicityBlack or African American	1.006	0.338	2.976	0.003	1.274
ethnicityNative Hawaiian or Pacific Islander	1.114	0.603	1.848	0.065	1.410
ethnicityOther	0.703	0.363	1.936	0.0533	0.890
ethnicityWhite	0.809	0.325	2.490	0.013	1.024
politicsIndependent	0.043	0.068	0.630	0.529	0.054
politicsRepublican	-0.077	0.073	-1.061	0.2892	-0.098
politicsSomething else	-0.099	0.152	-0.650	0.5161	-0.125
educationHigh school graduate	0.418	0.520	0.805	0.421	0.529
educationSome college	0.556	0.516	1.077	0.2819	0.704
education2 year degree	0.612	0.519	1.179	0.239	0.774
education4 year degree	0.553	0.515	1.074	0.2834	0.699
educationProfessional degree	0.444	0.517	0.858	0.391	0.562
educationDoctorate	0.605	0.562	1.077	0.2819	0.766
frame_attributePositive	0.352	0.194	1.817	0.0697	0.834
age	-0.005	0.003	-1.506	0.1327	-0.072
frame_attributePositive:age	0.008	0.005	1.658	0.0977	0.118

The frame × age interaction is not statistically discernible ($\hat{\beta}_3 = 0.008$, $p = 0.098$), and its effect size is very small.

Post hoc testing.

```
# Age slope within each frame: which frame drives any change?
emt_att <- emtrends(m_att, specs = "frame_attribute", var = "age")
test(emt_att) |> as.data.frame() |> kable(digits = 4)
```

frame_attribute	age.trend	SE	df	t.ratio	p.value
Negative	-0.0046	0.0030	658	-1.5055	0.1327
Positive	0.0029	0.0034	658	0.8632	0.3883

```
# Difference in slopes (= the interaction coefficient)
pairs(emt_att, reverse = TRUE) |> as.data.frame() |> kable(digits = 4)
```

contrast	estimate	SE	df	t.ratio	p.value
Positive - Negative	0.0075	0.0045	658	1.6585	0.0977

```
# Framing effect (Positive - Negative) at selected ages, with Cohen's d
emm_att <- emmeans(m_att, specs = "frame_attribute", by = "age",
                   at = list(age = c(20, 30, 40, 50, 60, 70)))
pairs(emm_att, reverse = TRUE) |>
  as.data.frame() |>
  mutate(d = estimate / sd(framing$resp_attribute)) |>
  kable(digits = 3)
```

contrast	age	estimate	SE	df	t.ratio	p.value	d
Positive - Negative	20	0.502	0.110	658	4.565	0	0.636
Positive - Negative	30	0.577	0.075	658	7.737	0	0.731
Positive - Negative	40	0.652	0.056	658	11.623	0	0.826
Positive - Negative	50	0.728	0.070	658	10.454	0	0.921
Positive - Negative	60	0.803	0.103	658	7.780	0	1.016
Positive - Negative	70	0.878	0.143	658	6.123	0	1.111

```
emm_plot <- function(mod, frame, ylab) {
  emmeans(mod, specs = frame, by = "age", at = list(age = 18:71),
          nuisance = c("gender", "ethnicity", "politics", "education")) |>
    as.data.frame() |>
    ggplot(aes(age, emmean, color = .data[[frame]], fill = .data[[frame]])) +
    geom_ribbon(aes(ymin = lower.CL, ymax = upper.CL), alpha = 0.2, color = NA) +
    geom_line(linewidth = 1) +
    labs(x = "Age (years)", y = ylab, color = "Frame", fill = "Frame") +
    theme_bw()
}
```

```
# Same model with the frame coded 0/1 so johnson_neyman() can treat it as the focal predictor
m_att_jn <- lm(resp_attribute ~ gender + ethnicity + politics + education + pos * age,
               data = framing |> mutate(pos = as.numeric(frame_attribute == "Positive")))
jn_att <- johnson_neyman(m_att_jn, pred = pos, modx = age, control.fdr = TRUE)
jn_att
```

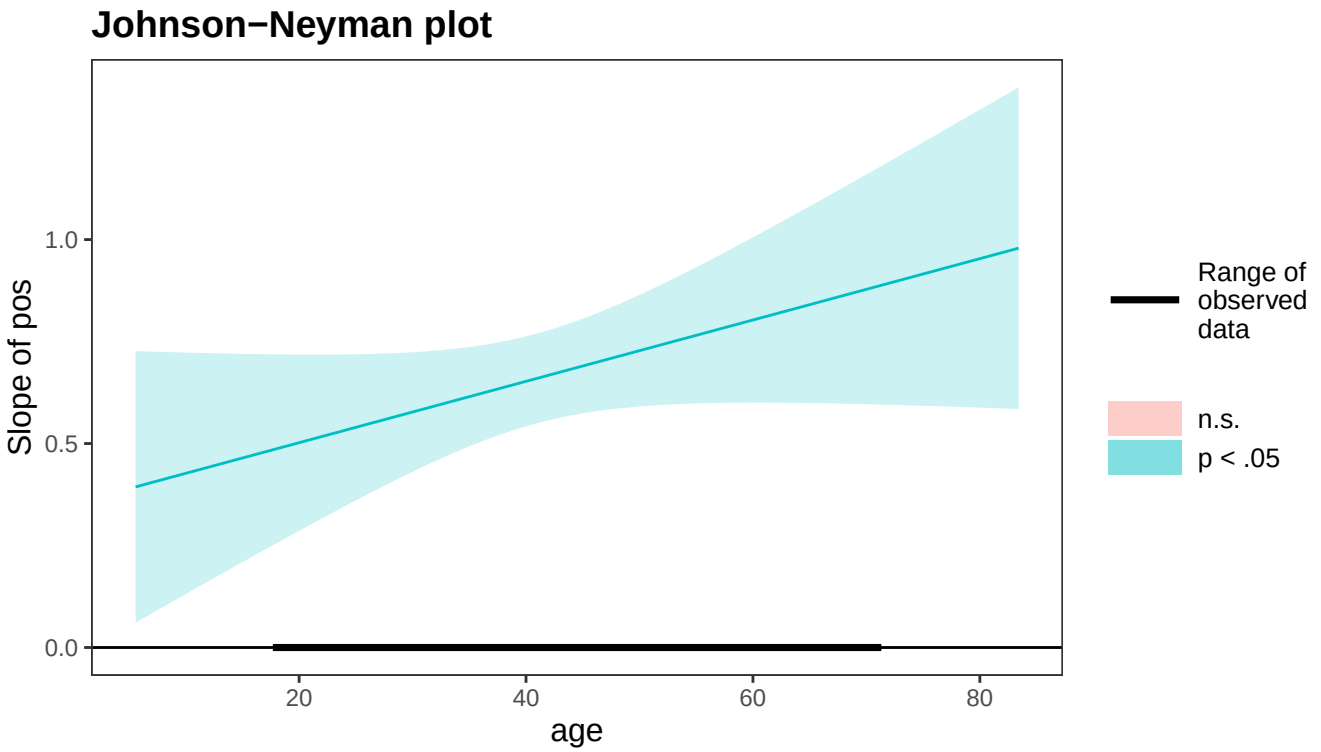
```
JOHNSON-NEYMAN INTERVAL

When age is INSIDE the interval [1.77, 516.35], the slope of pos is p <
.05.

Note: The range of observed values of age is [18.00, 71.00]

Interval calculated using false discovery rate adjusted t = 1.96
```

Figure 8: Johnson-Neyman plot: the framing effect (positive minus negative) across age, attribute framing.



Summary

- **Framing effect:** The positive frame ("70 felt better") produces higher efficacy ratings than the negative frame ("30 did not feel better") at every age from 18 to 71 (Figure 8). The effect is therefore present across the whole observed age range rather than "onsetting" at a particular age.
- **Change with age:** The estimated gap grows by about 0.08 points per decade, but this increase is not statistically discernible ($p = 0.098$).
- **Direction:** The negative-frame slope (-0.046 per decade, $p = 0.133$) and the positive-frame slope (0.029 per decade, $p = 0.388$) are both indistinguishable from zero. Descriptively, though, the widening comes from older adults rating the drug *lower* under the negative frame.

3.2 Likelihood to Stop Drinking Cocoa

```
m_goal <- lm(resp_goal ~ gender + ethnicity + politics + education +
               frame_goal * age,
             data = framing)
k_goal <- key_stats(m_goal, "frame_goal")
```

Conditions.

```
r <- resid_plots(m_goal); r$qq; r$rf
```

```
influence_plots(m_goal)
```

Figure 9: Residual diagnostics for the goal-framing model.

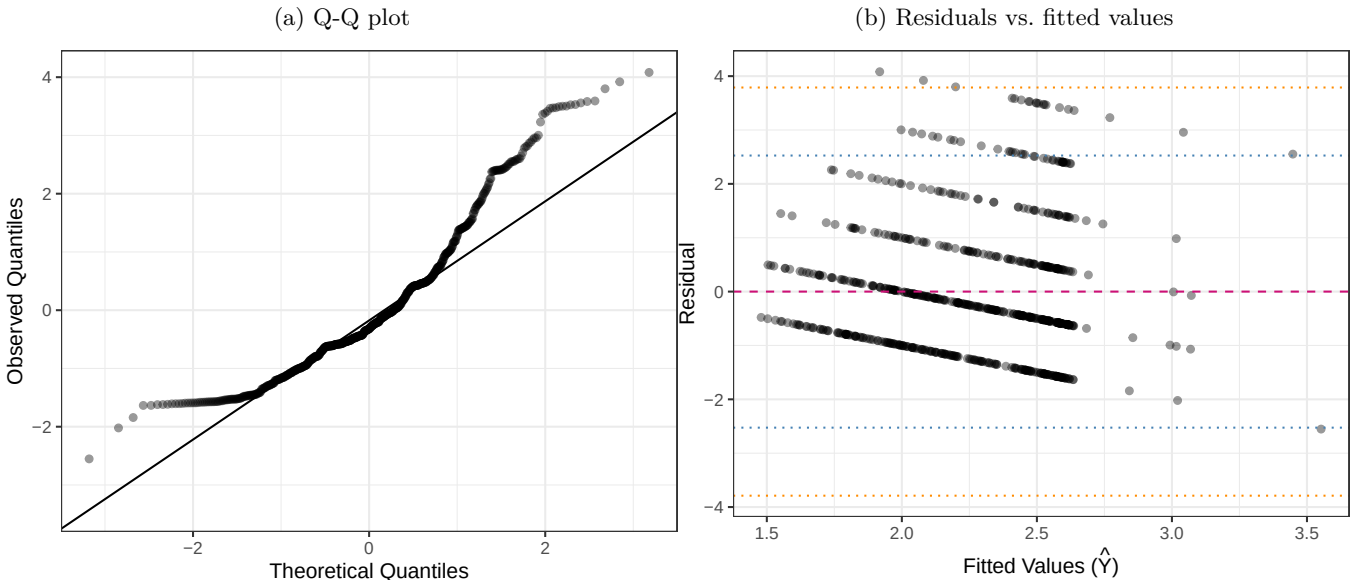
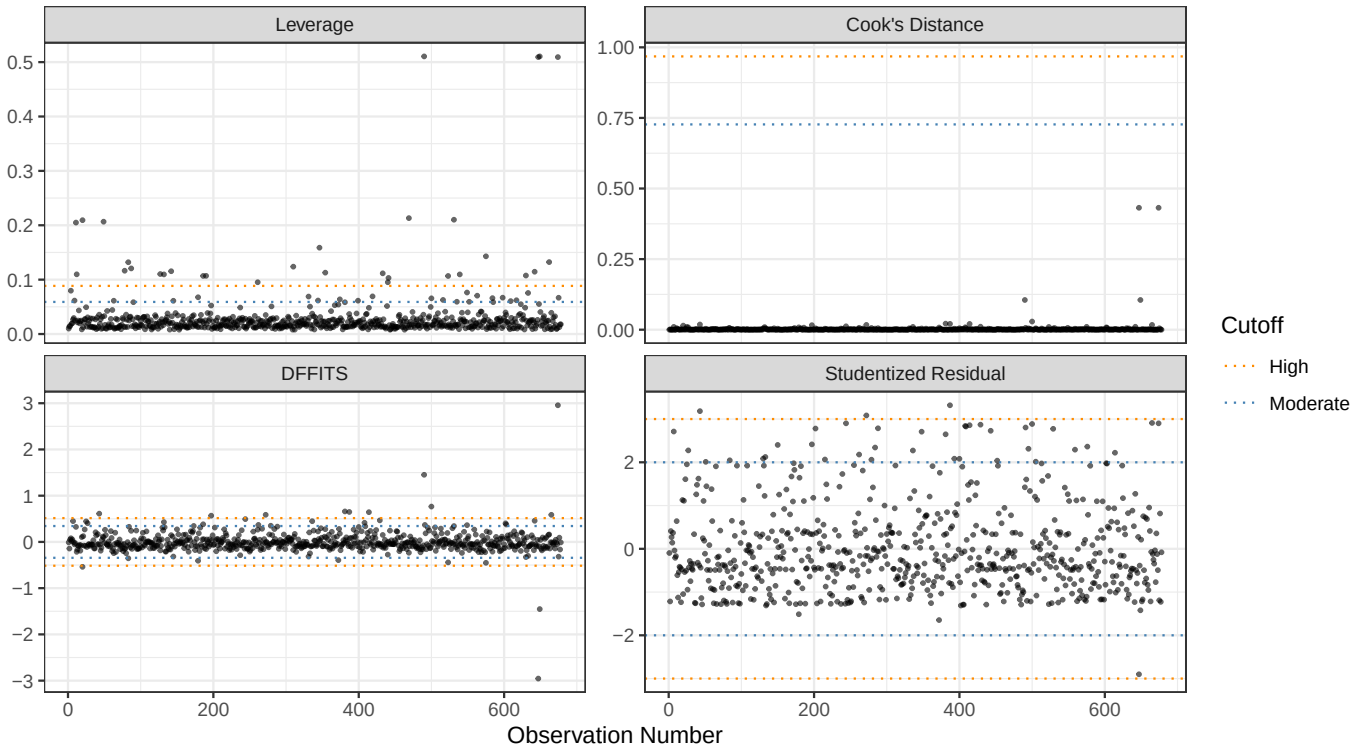


Figure 10: Influence diagnostics for the goal-framing model.




```
augment(m_goal) |> group_by(frame_goal) |> summarize(resid_sd = sd(.resid)) |> kable(digits = 3)
```

frame_goal	resid_sd
Negative	1.069
Positive	1.393

```r
vif(lm(resp_goal ~ gender + ethnicity + politics + education + frame_goal + age,
       data = framing))
```

```
               GVIF Df GVIF^(1/(2*Df))
gender     1.068292  2        1.016653
ethnicity  1.154549  5        1.014475
politics   1.176864  3        1.027514
education  1.178615  6        1.013789
frame_goal 1.033384  1        1.016555
age        1.060030  1        1.029578
```

```r
bc <- MASS::boxcox(m_goal, plotit = FALSE); bc$x[which.max(bc$y)]
```

```
[1] -0.2
```

- **Gaussian errors:** The errors might not be Gaussian. Most participants chose 1 or 2, so the residuals are right-skewed, and the Q-Q plot bends upward in the tail (Figure 9).
- **Constant variance:** The residual spread is larger under the positive frame.
- **Box-Cox:** It suggests $\lambda \approx 0$, i.e., a log transformation. Following the lecture's approach, we refit the model on the log scale as a robustness check below, but keep the untransformed model as the primary analysis because it can be interpreted on the rating scale.
- **Influence and multicollinearity:** No observation is influential, and all VIFs are close to 1.

Overall F-test and effect size.

```r
anova(m_goal) |> as.data.frame() |> kable(digits = 3)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
gender	2	0.792	0.396	0.248	0.780
ethnicity	5	4.882	0.976	0.612	0.691
politics	3	4.477	1.492	0.936	0.423
education	6	9.897	1.650	1.034	0.402
frame_goal	1	41.488	41.488	26.011	0.000
age	1	3.533	3.533	2.215	0.137
frame_goal:age	1	4.555	4.555	2.856	0.092
Residuals	658	1049.521	1.595	NA	NA

```r
F_to_omega2(f = k_goal$F, df = k_goal$df1, df_error = k_goal$df2)
```

```
Omega2 (partial) |       95% CI
--------------------------------
0.04             | [0.00, 1.00]

- One-sided CIs: upper bound fixed at [1.00].
```

```r
interpret_omega_squared(F_to_omega2(k_goal$F, k_goal$df1, k_goal$df2)$Omega2_partial, rules = "cohen1992")
```

```
[1] "small"
(Rules: cohen1992)
```

The model is statistically discernible and explains a small portion of the variation ($R^2 = 0.062$, adjusted $R^2 = 0.035$; $F(19, 658) = 2.3$, p = 0.001).

```
term_effects(m_goal)
```

Table 4: Effect size (partial omega-squared) for each term, goal framing.

Term	Partial omega^2	Lower	Upper	Size
gender	0.000	0.000	1	very small
ethnicity	0.000	0.000	1	very small
politics	0.000	0.000	1	very small
education	0.000	0.000	1	very small
frame_goal	0.036	0.016	1	small
age	0.002	0.000	1	very small
frame_goal:age	0.003	0.000	1	very small

```
coef_table(m_goal)
```

Table 5: Coefficients for the goal-framing model.

Term	Estimate	SE	t	p	Std-Coef
(Intercept)	2.518	1.096	2.298	0.0219	-0.210
genderMale	0.031	0.101	0.311	0.756	0.024
genderNon-binary / third gender	0.099	0.411	0.241	0.8098	0.077
ethnicityAsian	0.315	0.594	0.531	0.5955	0.245
ethnicityBlack or African American	0.177	0.595	0.297	0.7663	0.138
ethnicityNative Hawaiian or Pacific Islander	1.230	1.060	1.160	0.2465	0.957
ethnicityOther	0.276	0.641	0.431	0.6667	0.215
ethnicityWhite	0.301	0.572	0.526	0.5988	0.234
politicsIndependent	-0.052	0.120	-0.431	0.6664	-0.040
politicsRepublican	-0.001	0.128	-0.006	0.9953	-0.001
politicsSomething else	0.447	0.268	1.668	0.0957	0.348
educationHigh school graduate	-0.480	0.914	-0.525	0.5997	-0.373
educationSome college	-0.279	0.907	-0.307	0.7587	-0.217
education2 year degree	-0.400	0.913	-0.438	0.6617	-0.311
education4 year degree	-0.221	0.904	-0.244	0.8072	-0.172
educationProfessional degree	-0.358	0.909	-0.394	0.6935	-0.279
educationDoctorate	-0.304	0.987	-0.308	0.758	-0.237
frame_goalPositive	-0.042	0.338	-0.124	0.9011	0.393
age	-0.013	0.006	-2.246	0.025	-0.123
frame_goalPositive:age	0.013	0.008	1.690	0.0915	0.129

Post hoc testing.

```
emt_goal <- emtrends(m_goal, specs = "frame_goal", var = "age")
test(emt_goal) |> as.data.frame() |> kable(digits = 4)
```

frame_goal	age.trend	SE	df	t.ratio	p.value
Negative	-0.0127	0.0057	658	-2.2461	0.0250
Positive	0.0007	0.0056	658	0.1228	0.9023

```r
pairs(emt_goal, reverse = TRUE) |> as.data.frame() |> kable(digits = 4)
```

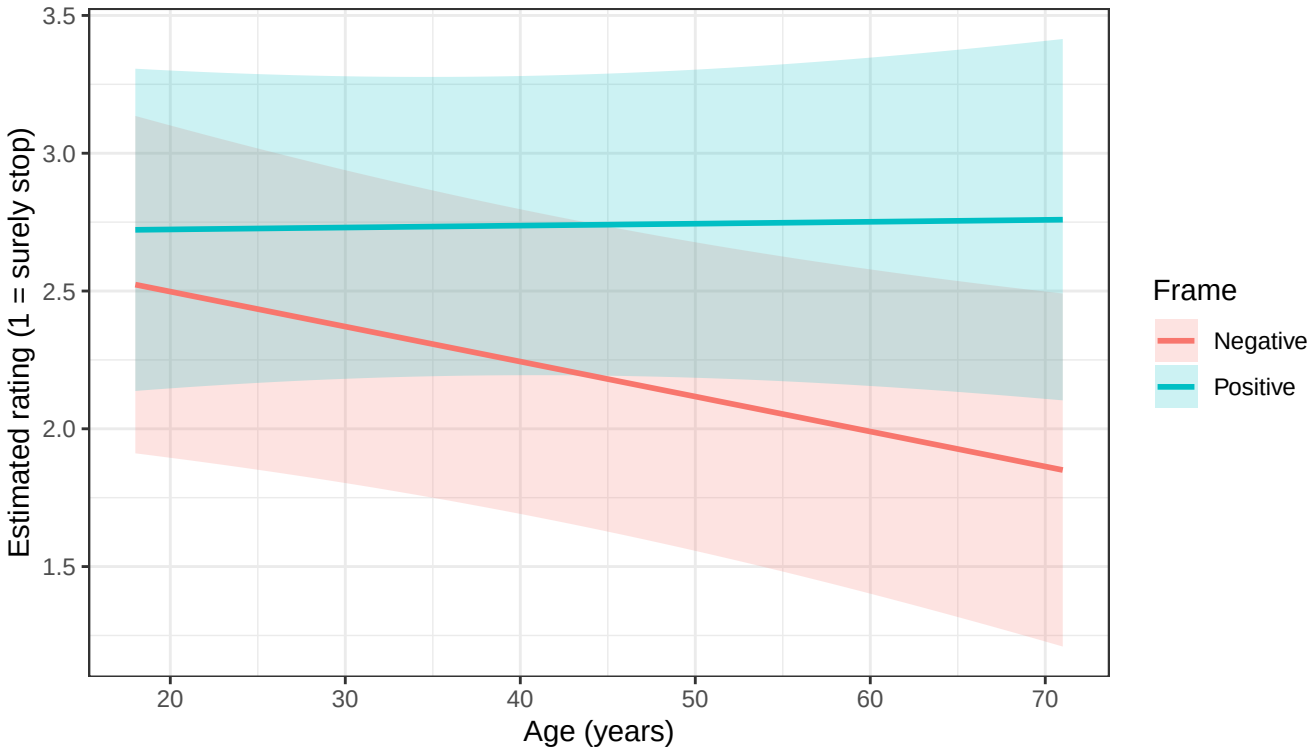
contrast	estimate	SE	df	t.ratio	p.value
Positive - Negative	0.0134	0.0079	658	1.6899	0.0915

```r
emm_goal <- emmeans(m_goal, specs = "frame_goal", by = "age",
                    at = list(age = c(20, 30, 40, 50, 60, 70)))
pairs(emm_goal, reverse = TRUE) |>
  as.data.frame() |>
  mutate(d = estimate / sd(framing$resp_goal)) |>
  kable(digits = 3)
```

contrast	age	estimate	SE	df	t.ratio	p.value	d
Positive - Negative	20	0.226	0.192	658	1.172	0.241	0.176
Positive - Negative	30	0.360	0.131	658	2.746	0.006	0.280
Positive - Negative	40	0.493	0.099	658	4.989	0.000	0.384
Positive - Negative	50	0.627	0.122	658	5.126	0.000	0.488
Positive - Negative	60	0.761	0.181	658	4.208	0.000	0.592
Positive - Negative	70	0.895	0.251	658	3.565	0.000	0.696

```r
emm_plot(m_goal, "frame_goal", "Estimated rating (1 = surely stop)")
```

Figure 11: Estimated marginal mean likelihood of continuing to drink cocoa by frame across age (95% CI).




```r
m_goal_jn <- lm(resp_goal ~ gender + ethnicity + politics + education + pos * age,
                data = framing |> mutate(pos = as.numeric(frame_goal == "Positive")))
jn_goal <- johnson_neyman(m_goal_jn, pred = pos, modx = age, control.fdr = TRUE)
jn_goal
```

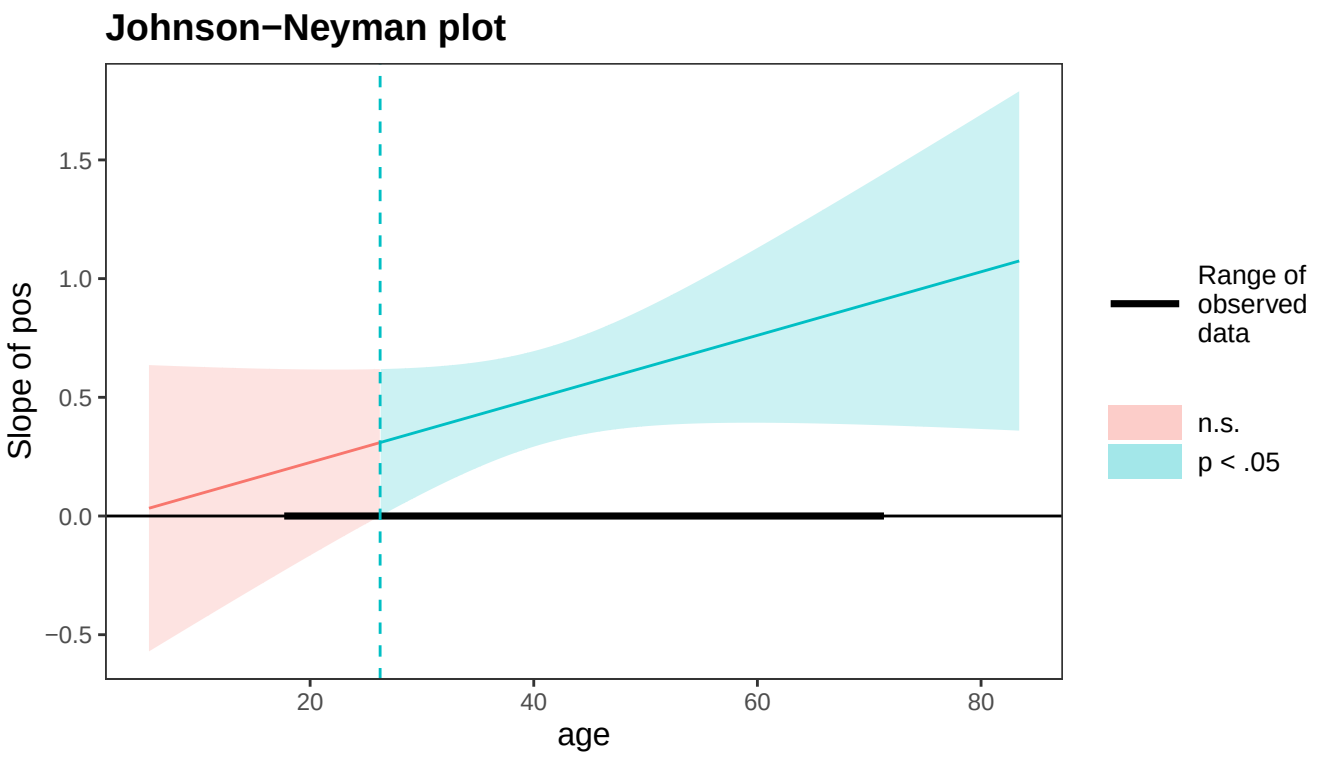
```
JOHNSON-NEYMAN INTERVAL

When age is INSIDE the interval [26.26, 222.81], the slope of pos is p <
.05.

Note: The range of observed values of age is [18.00, 71.00]

Interval calculated using false discovery rate adjusted t = 2.04
```

Figure 12: Johnson-Neyman plot: the framing effect (positive minus negative) across age, goal framing.



Summary.

- **Change with age:** The interaction is in the predicted direction ($\hat{\beta}_3 = 0.0134$), but it is not statistically discernible at $\alpha = 0.05$ (p = 0.092), and its effect size is very small.
- **Direction:** The pattern is driven by the negative frame. Under "continuing increases cancer risk," older participants were more likely to say they would stop drinking cocoa (slope = -0.127 per decade, p = 0.025). Under the positive frame, age had essentially no relationship with the rating (0.007 per decade, p = 0.902).
- **Onset:** The Johnson-Neyman analysis shows that the framing effect is not discernible among the youngest adults. It becomes statistically discernible at about age 27 and stays discernible through age 71 (Figure 12).

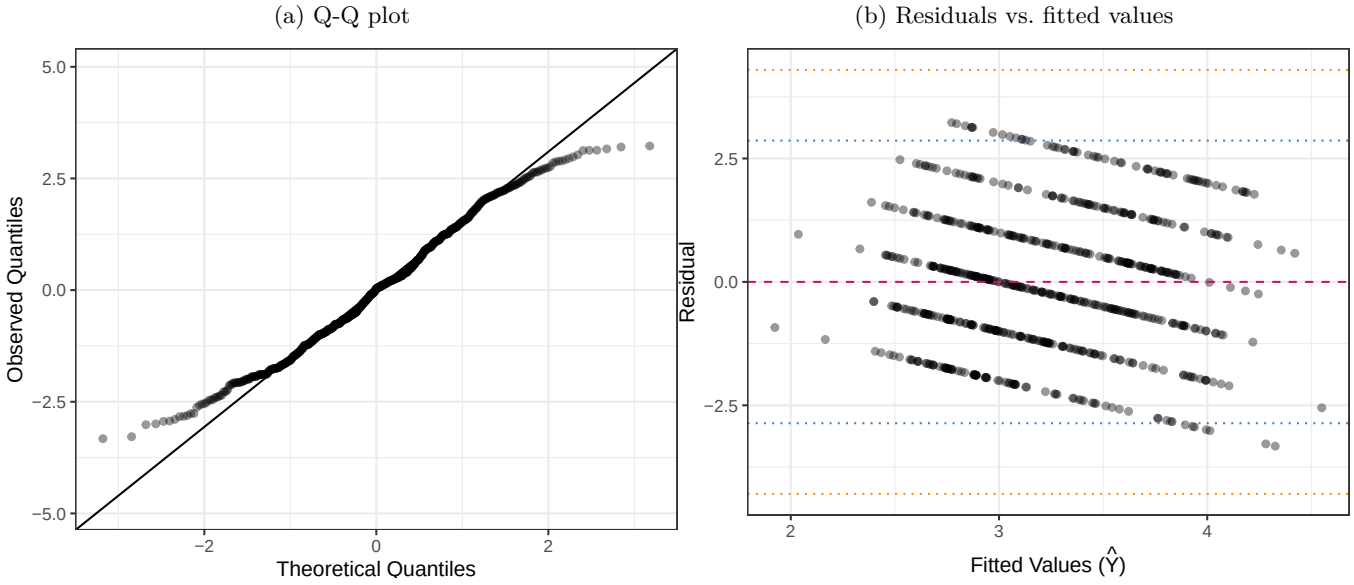
3.3 Preference for Chance Choice

```r
m_risky <- lm(resp_risky ~ gender + ethnicity + politics + education +
                frame_risky * age,
              data = framing)
k_risky <- key_stats(m_risky, "frame_risky")
```

Conditions.

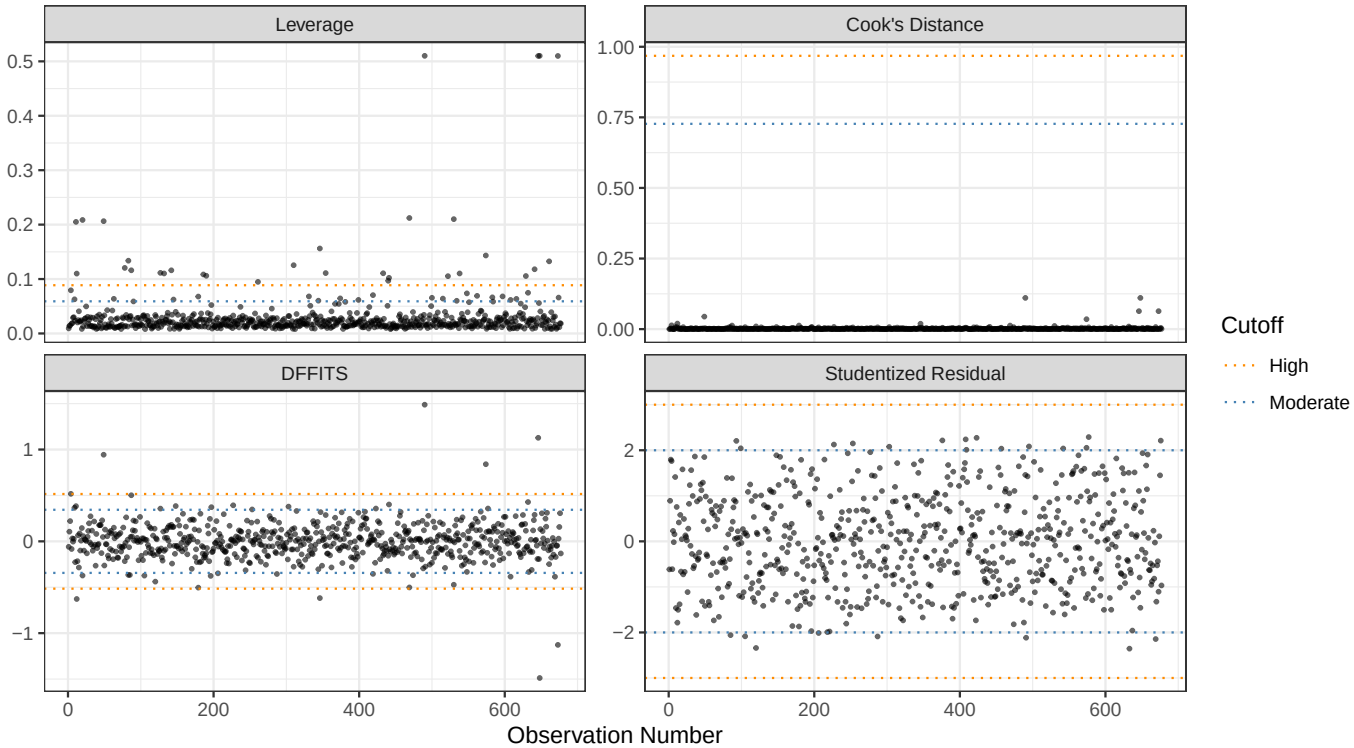
```
r <- resid_plots(m_risky); r$qq; r$rf
```

Figure 13: Residual diagnostics for the risky-choice model.




```
influence_plots(m_risky)
```

Figure 14: Influence diagnostics for the risky-choice model.




```
augment(m_risky) |> group_by(frame_risky) |> summarize(resid_sd = sd(.resid)) |> kable(digits = 3)
```

frame_risky	resid_sd
Negative	1.471
Positive	1.351

```
vif(lm(resp_risky ~ gender + ethnicity + politics + education + frame_risky + age,
       data = framing))
```

```
                GVIF Df GVIF^(1/(2*Df))
gender      1.065405  2        1.015965
ethnicity   1.150583  5        1.014126
politics    1.184665  3        1.028646
education   1.158332  6        1.012324
frame_risky 1.026221  1        1.013026
age         1.068738  1        1.033798
```

```
bc <- MASS::boxcox(m_risky, plotit = FALSE); bc$x[which.max(bc$y)]
```

```
[1] 0.6
```

- **Gaussian errors:** Ratings use the whole 1–6 scale, so the residuals are roughly symmetric. The tails are slightly lighter than Gaussian because the scale is bounded (Figure 13).
- **Constant variance:** The residual spread is similar under the two frames.
- **Box-Cox:** λ is between 0.5 and 1, so no transformation is needed.
- **Influence and multicollinearity:** There are no influential points, and all VIFs are close to 1.

Overall F-test and effect size.

```
anova(m_risky) |> as.data.frame() |> kable(digits = 3)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
gender	2	1.919	0.960	0.469	0.626
ethnicity	5	12.048	2.410	1.176	0.319
politics	3	32.900	10.967	5.354	0.001
education	6	10.076	1.679	0.820	0.555
frame_risky	1	71.789	71.789	35.045	0.000
age	1	6.386	6.386	3.118	0.078
frame_risky:age	1	10.274	10.274	5.016	0.025
Residuals	657	1345.842	2.048	NA	NA

```
F_to_omega2(f = k_risky$F, df = k_risky$df1, df_error = k_risky$df2)
```

```
Omega2 (partial) |       95% CI
--------------------------------
0.07             | [0.02, 1.00]

- One-sided CIs: upper bound fixed at [1.00].
```

```
interpret_omega_squared(F_to_omega2(k_risky$F, k_risky$df1, k_risky$df2)$Omega2_partial, rules = "cohen1992
```

```
[1] "small"
(Rules: cohen1992)
```

The model is statistically discernible ($R^2 = 0.097$, adjusted $R^2 = 0.071$; $F(19, 657) = 3.74$, $p < 0.001$).

```
term_effects(m_risky)
```

Table 6: Effect size (partial omega-squared) for each term, risky-choice framing.

Term	Partial omega^2	Lower	Upper	Size
gender	0.000	0.000	1	very small
ethnicity	0.001	0.000	1	very small
politics	0.019	0.003	1	very small
education	0.000	0.000	1	very small
frame_risky	0.048	0.025	1	small
age	0.003	0.000	1	very small
frame_risky:age	0.006	0.000	1	very small

```
coef_table(m_risky)
```

Table 7: Coefficients for the risky-choice model.

Term	Estimate	SE	t	p	Std-Coef
(Intercept)	1.212	1.244	0.975	0.33	-0.872
genderMale	0.136	0.115	1.190	0.2345	0.092
genderNon-binary / third gender	-0.111	0.464	-0.240	0.8104	-0.075
ethnicityAsian	0.893	0.672	1.329	0.1844	0.601
ethnicityBlack or African American	1.031	0.675	1.528	0.1269	0.694
ethnicityNative Hawaiian or Pacific Islander	0.896	1.200	0.746	0.4557	0.603
ethnicityOther	0.668	0.726	0.920	0.358	0.450
ethnicityWhite	0.716	0.648	1.105	0.2696	0.482
politicsIndependent	0.489	0.137	3.564	4e-04	0.329
politicsRepublican	0.175	0.145	1.206	0.2281	0.118
politicsSomething else	0.089	0.304	0.294	0.7689	0.060
educationHigh school graduate	0.618	1.036	0.597	0.5506	0.416
educationSome college	0.708	1.028	0.689	0.4913	0.477
education2 year degree	0.431	1.034	0.416	0.6772	0.290
education4 year degree	0.717	1.025	0.700	0.4843	0.483
educationProfessional degree	0.477	1.030	0.464	0.643	0.321
educationDoctorate	0.369	1.119	0.329	0.742	0.248
frame_riskyPositive	0.188	0.386	0.487	0.6262	-0.430
age	0.018	0.006	2.824	0.0049	0.150
frame_riskyPositive:age	-0.020	0.009	-2.240	0.0255	-0.169

Post hoc testing.

```
emt_risky <- emtrends(m_risky, specs = "frame_risky", var = "age")
test(emt_risky) |> as.data.frame() |> kable(digits = 4)
```

frame_risky	age.trend	SE	df	t.ratio	p.value
Negative	0.0179	0.0063	657	2.8237	0.0049
Positive	-0.0023	0.0065	657	-0.3551	0.7226

```r
pairs(emt_risky, reverse = TRUE) |> as.data.frame() |> kable(digits = 4)
```

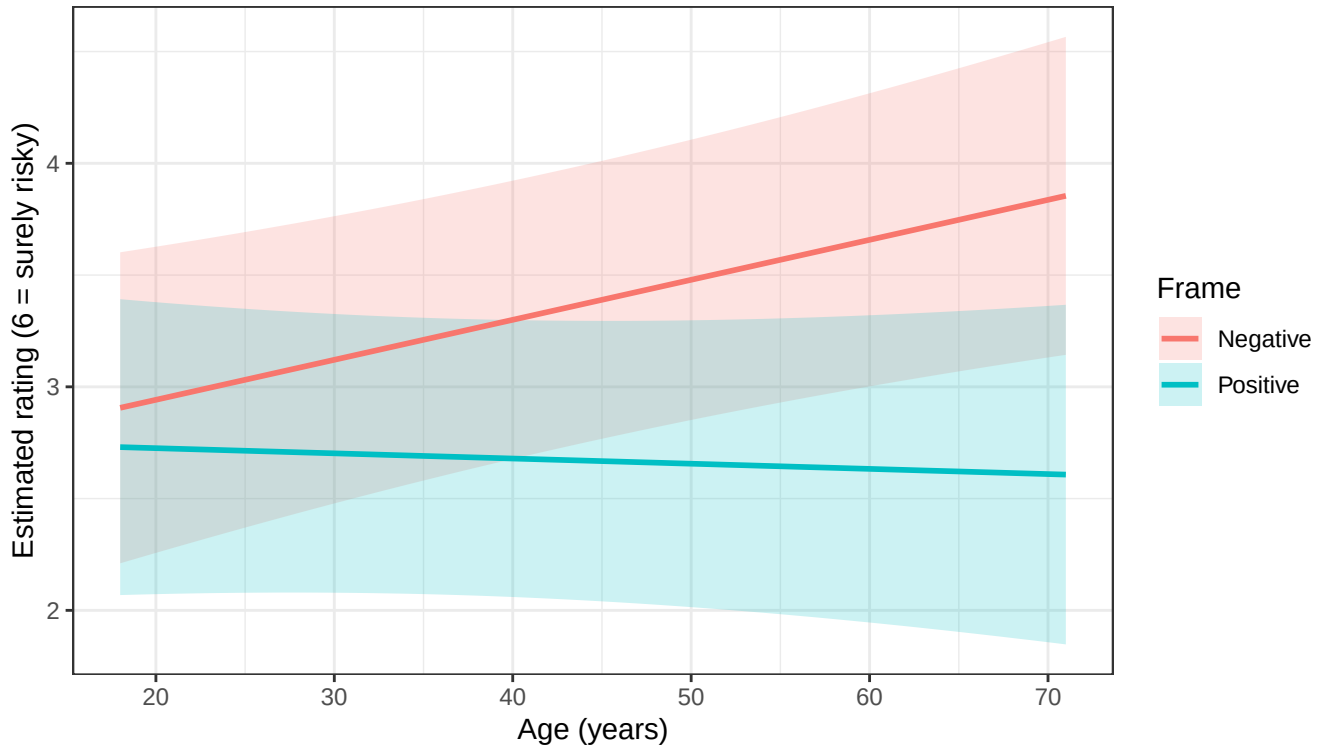
contrast	estimate	SE	df	t.ratio	p.value
Positive - Negative	-0.0202	0.009	657	-2.2396	0.0255

```r
emm_risky <- emmeans(m_risky, specs = "frame_risky", by = "age",
                     at = list(age = c(20, 30, 40, 50, 60, 70)))
pairs(emm_risky, reverse = TRUE) |>
  as.data.frame() |>
  mutate(d = estimate / sd(framing$resp_risky, na.rm = TRUE)) |>
  kable(digits = 3)
```

contrast	age	estimate	SE	df	t.ratio	p.value	d
Positive - Negative	20	-0.216	0.219	657	-0.987	0.324	-0.146
Positive - Negative	30	-0.418	0.149	657	-2.815	0.005	-0.282
Positive - Negative	40	-0.621	0.112	657	-5.553	0.000	-0.418
Positive - Negative	50	-0.823	0.138	657	-5.943	0.000	-0.554
Positive - Negative	60	-1.025	0.205	657	-4.993	0.000	-0.690
Positive - Negative	70	-1.227	0.285	657	-4.301	0.000	-0.826

```r
emm_plot(m_risky, "frame_risky", "Estimated rating (6 = surely risky)")
```

Figure 15: Estimated marginal mean preference for the risky option by frame across age (95% CI).




```r
m_risky_jn <- lm(resp_risky ~ gender + ethnicity + politics + education + pos * age,
                 data = framing |> mutate(pos = as.numeric(frame_risky == "Positive")))
jn_risky <- johnson_neyman(m_risky_jn, pred = pos, modx = age, control.fdr = TRUE)
jn_risky
```

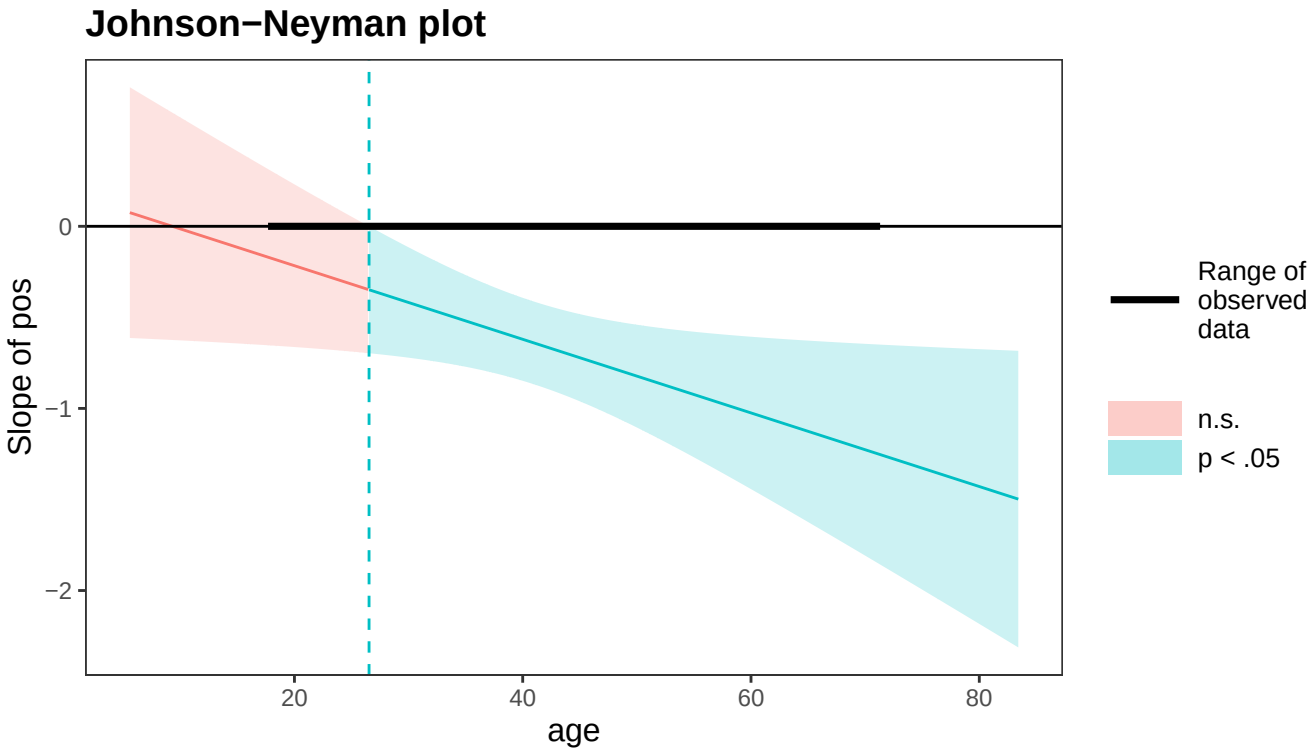
```
JOHNSON-NEYMAN INTERVAL

When age is OUTSIDE the interval [-313.21, 26.54], the slope of pos is p <
.05.

Note: The range of observed values of age is [18.00, 71.00]

Interval calculated using false discovery rate adjusted t = 2.04
```

Figure 16: Johnson-Neyman plot: the framing effect (positive minus negative) across age, risky-choice framing.



Summary. Risky choice is the only scenario with a statistically discernible frame × age interaction ($\hat{\beta}_3$ = -0.0202, p = 0.025), although its effect size is very small.

- **Direction:** The change is driven entirely by the negative frame. When the options were described in terms of deaths, preference for the risky program rose with age (0.179 points per decade, p = 0.005). When they were described in terms of lives saved, age had no relationship with preference (-0.023 per decade, p = 0.723).
- **Onset:** The framing effect is not discernible among the youngest participants. It becomes statistically discernible at about age 26.5 (Figure 16).

4 Discussion

References

Xu, Lance. 2024. "Advanced Age Predicts Increased Susceptibility to Attribute, Goal, and Risky-Choice Framing in Negative Frame Valences." *Humanities and Social Sciences Communications* 11 (1): 217.